

1. Introduction to Databases

What is a Database?

- Organized collection of data.
- Used to store, retrieve, update, and manage information.

Why Databases?

- Faster searching.
- Data consistency.
- Multiple users can access data.
- Easier data management.

Types

- Relational Databases (MySQL, PostgreSQL, SQL Server)
- Non-Relational Databases (MongoDB)

Key Terms

- Database → Container
- Table → Stores data
- Row → Record
- Column → Attribute

Example:

Students

```
-----  
id | name  | age  
-----
```

```
1 | John  | 20
```

```
2 | Emma  | 22
```

2. Installing MySQL & Basic Setup

Connect to MySQL

```
mysql -u root -p
```

Show Databases

```
SHOW DATABASES;
```

Create Database

```
CREATE DATABASE shop;
```

Use Database

```
USE shop;
```

Delete Database

```
DROP DATABASE shop;
```

Remember:

- **CREATE** = Create
 - **USE** = Select
 - **DROP** = Delete permanently
-

3. Creating Tables

Create Table

```
CREATE TABLE cats(  
    name VARCHAR(50),  
    age INT  
);
```

Show Tables

```
SHOW TABLES;
```

Describe Table

```
DESC cats;
```

Delete Table

```
DROP TABLE cats;
```

4. Inserting Data

Insert One Row

```
INSERT INTO cats(name, age)
VALUES('Blue', 3);
```

Insert Multiple Rows

```
INSERT INTO cats(name, age)
VALUES
('Tom',2),
('Kitty',4),
('Leo',1);
```

View Data

```
SELECT * FROM cats;
```

Most common command:

```
SELECT * FROM table_name;
```

5. CRUD Operations

CRUD = Create, Read, Update, Delete

READ

```
SELECT * FROM cats;
```

Specific Columns:

```
SELECT name FROM cats;
```

Multiple Columns:

```
SELECT name, age FROM cats;
```

Alias:

```
SELECT name AS Cat_Name  
FROM cats;
```

WHERE

Filter data.

```
SELECT *  
FROM cats  
WHERE age = 4;
```

```
SELECT *  
FROM cats  
WHERE name='Tom';
```

UPDATE

```
UPDATE cats  
SET age=5  
WHERE name='Tom';
```

Always use WHERE.

DELETE

```
DELETE FROM cats  
WHERE age=1;
```

Delete all rows:

```
DELETE FROM cats;
```

Delete table:

```
DROP TABLE cats;
```

6. String Functions

CONCAT

```
SELECT CONCAT(first_name,',',last_name)  
FROM users;
```

Result:

John Doe

CONCAT_WS

Separator automatically added.

```
SELECT CONCAT_WS('-',day,month,year);
```

Output:

10-04-2025

SUBSTRING

```
SELECT SUBSTRING(title,1,5);
```

Output:

Harry

REPLACE

```
SELECT REPLACE('Hello World','World','SQL');
```

Output:

Hello SQL

REVERSE

```
SELECT REVERSE('Hello');
```

Output:

olleH

CHAR_LENGTH

```
SELECT CHAR_LENGTH('Hello');
```

Output:

5

UPPER / LOWER

```
SELECT UPPER(name);  
SELECT LOWER(name);
```

7. Refining Selections

DISTINCT

Remove duplicates.

```
SELECT DISTINCT author_fname  
FROM books;
```

ORDER BY

Ascending:

```
SELECT * FROM books  
ORDER BY released_year;
```

Descending:

```
SELECT * FROM books  
ORDER BY released_year DESC;
```

LIMIT

```
SELECT * FROM books  
LIMIT 5;
```

Pagination:

LIMIT 0,5

LIMIT 5,5

LIKE

Wildcard search.

%

Any number of characters.

WHERE title LIKE '%story%'

_

Single character.

WHERE name LIKE '_a%'

8. Aggregate Functions

Used for calculations.

COUNT

```
SELECT COUNT(*)  
FROM books;
```

COUNT DISTINCT

```
SELECT COUNT(DISTINCT author_fname)  
FROM books;
```

MIN

```
SELECT MIN(released_year)
FROM books;
```

MAX

```
SELECT MAX(pages)
FROM books;
```

SUM

```
SELECT SUM(pages)
FROM books;
```

AVG

```
SELECT AVG(pages)
FROM books;
```

9. GROUP BY

Most important SQL concept.

Groups rows before aggregation.

```
SELECT author_fname,
COUNT(*)
FROM books
GROUP BY author_fname;
```

Output:

John 5
Mark 3

Average by group:

```
SELECT author_fname,  
AVG(pages)  
FROM books  
GROUP BY author_fname;
```

10. Data Types

Numeric

INT
BIGINT
DECIMAL
FLOAT
DOUBLE

Money:

DECIMAL(10,2)

Preferred over FLOAT.

String

CHAR
VARCHAR
TEXT

Most used:

VARCHAR(255)

Date & Time

DATE

TIME

DATETIME

TIMESTAMP

Current Date:

CURDATE();

Current Time:

CURTIME();

Current DateTime:

NOW();

11. Logical Operators

AND

WHERE age > 18

AND city='Delhi'

OR

WHERE city='Delhi'

OR city='Mumbai'

BETWEEN

WHERE age BETWEEN 18 AND 30

IN

WHERE city IN ('Delhi','Mumbai')

NOT IN

WHERE city NOT IN ('Delhi','Mumbai')

CASE

SQL if-else.

```
SELECT title,  
CASE  
  WHEN stock > 50 THEN 'Available'  
  ELSE 'Low Stock'  
END  
FROM books;
```

12. One-to-Many Relationships

Example:

Customers
Orders

One customer → Many orders

Customers

customer_id
name

Orders

order_id
amount
customer_id

Foreign Key:

FOREIGN KEY(customer_id)
REFERENCES customers(id)

13. JOINS

Most important section.

INNER JOIN

Matching rows only.

```
SELECT *  
FROM customers  
JOIN orders  
ON customers.id=orders.customer_id;
```

LEFT JOIN

All left table rows.

```
SELECT *  
FROM customers  
LEFT JOIN orders  
ON customers.id=orders.customer_id;
```

RIGHT JOIN

All right table rows.

```
SELECT *  
FROM customers  
RIGHT JOIN orders  
ON customers.id=orders.customer_id;
```

14. Many-to-Many Relationships

Example:

Students ↔ Courses

Need Junction Table.

```
students  
courses  
student_course
```

Example Structure

```
students(id,name)
```

```
courses(id,title)
```

```
student_course(  
  student_id,  
  course_id  
)
```

15. Instagram Clone Project

Tables commonly created:

```
users
```

photos
comments
likes
follows
tags
photo_tags

Concepts learned:

- Database Design
- Relationships
- Foreign Keys
- Real-world Queries

Example:

Most liked photo:

```
SELECT photo_id,  
COUNT(*) AS likes  
FROM likes  
GROUP BY photo_id  
ORDER BY likes DESC  
LIMIT 1;
```

16. Database Constraints

NOT NULL

name VARCHAR(50) NOT NULL

DEFAULT

age INT DEFAULT 18

UNIQUE

email VARCHAR(255) UNIQUE

PRIMARY KEY

id INT PRIMARY KEY

AUTO_INCREMENT

id INT AUTO_INCREMENT PRIMARY KEY

Most common:

id INT AUTO_INCREMENT PRIMARY KEY

17. Final Must-Know Commands

Database

CREATE DATABASE
DROP DATABASE
USE database
SHOW DATABASES

Tables

CREATE TABLE
ALTER TABLE
DROP TABLE
DESC table

Data

INSERT
SELECT

UPDATE
DELETE

Filtering

WHERE
LIKE
IN
BETWEEN
ORDER BY
LIMIT
DISTINCT

Aggregation

COUNT
SUM
AVG
MIN
MAX
GROUP BY

Relationships

PRIMARY KEY
FOREIGN KEY
JOIN
LEFT JOIN
RIGHT JOIN
